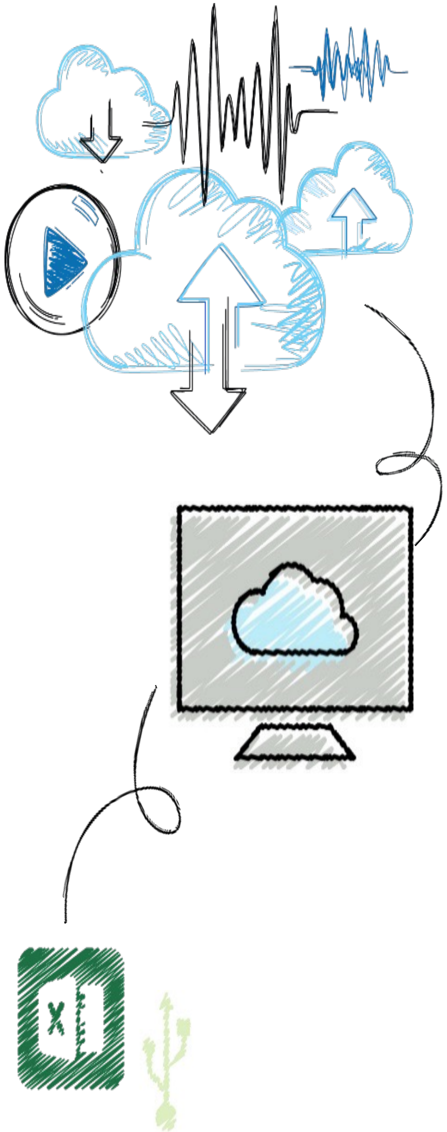
ICT703 Programming

Module 3 Topic 2

Selection

CENGAGE



Objectives

1. Use selection statements to make choices in a program
2. Construct appropriate conditions for condition-controlled loops and selection statements
3. Use a selection statement and a break statement to exit a loop that is not entry-controlled

Programming Concepts

Four control flow

- Sequence
- Iteration
- Selection
- Abstraction – methods/classes

- Plus Data

Selection: If and If-Else Statements

- **Selection statements** allow a computer to make choices
  - Based on a **condition**

One-Way Selection Statements

- Simplest form of selection is the **if statement**

if <condition>:

<sequence of statements>

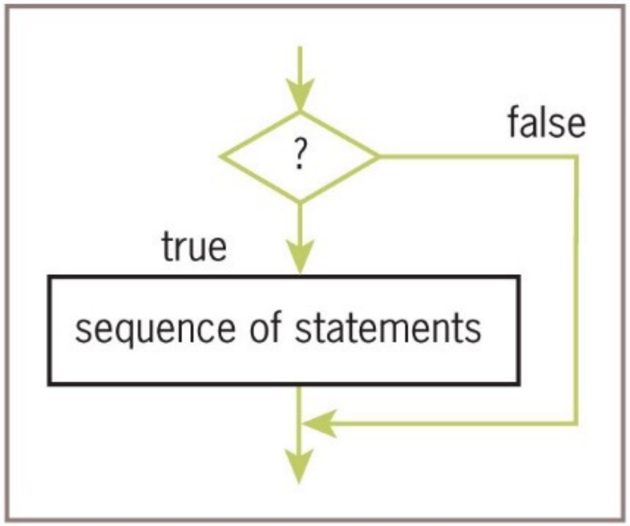


Figure 3-3 The semantics of the `if` statement

One-Way Selection Example

- **Problem**: Do I need an umbrella?

- **Pseudo Code**:

If it is raining then take an umbrella

- **Code**

```
weather = input("Is it raining: ")
If weather == "yes":
        print("Take an umbrella")
```

Two way If-Else Statements

Syntax

if <condition>:

<sequence of statements–1>

else:

<sequence of statements-2>

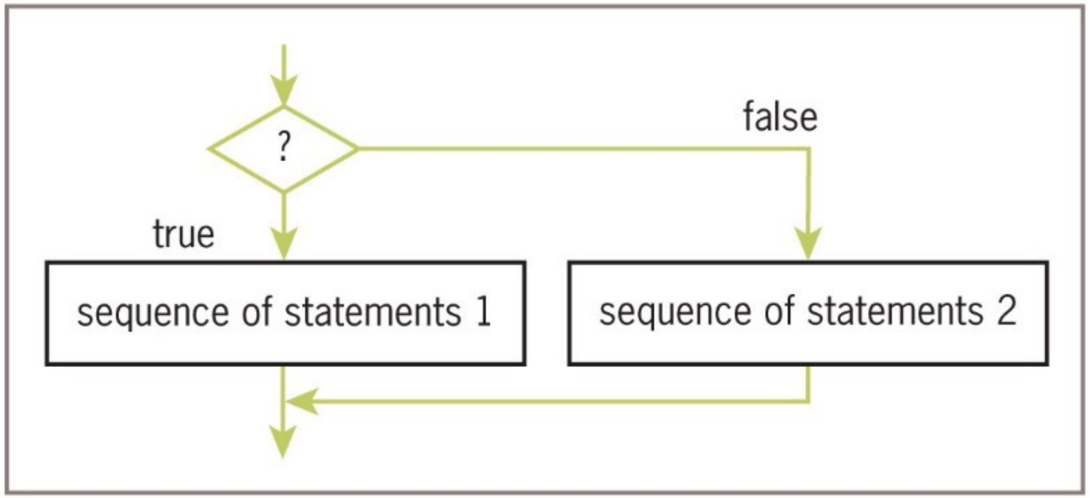


Figure 3-2 The semantics of the **`if-else`** statement

Two-Way Selection Example

- **Problem**: What do I need?
- **Pseudo Code**:

If it is raining then take an umbrella

Otherwise take sunnies

- **Code**

```
weather = input("Is it raining: ")
If weather == "yes":
        print("Take an umbrella")
else:
        print("Take sunnies")
```

Multi-Way If Statements

Syntax:
if <condition-1>:
<sequence of statements-1>
elif <condition-*n*>:
<sequence of statements-*n*>
else:
<default sequence of statements>

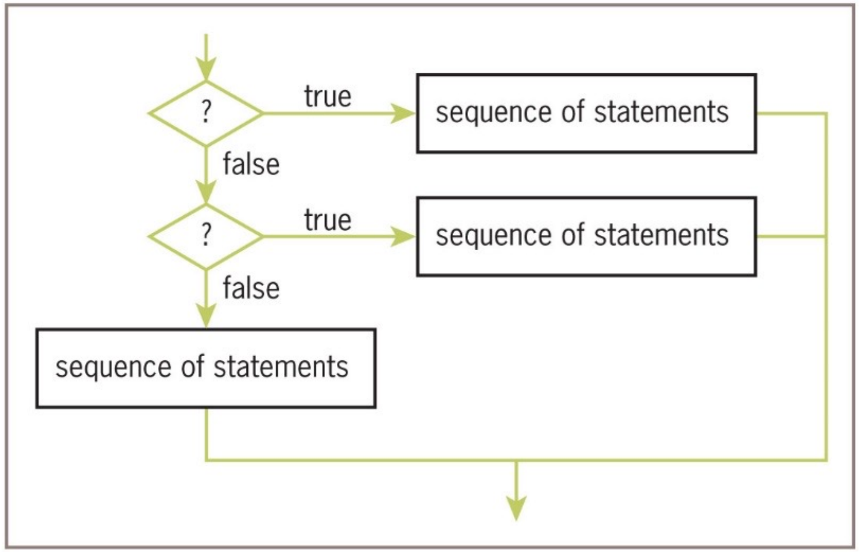


Figure 3-4 The semantics of the multi-way `if` statement

Multi-Way If Statements Example

- A program may be faced with testing conditions that entail more than two alternative courses of action
- Can be described in code by a **multi-way selection statement**

Letter Grade	**Range of Numeric Grades**
A	All grades above 89
B	All grades above 79 and below 90
C	All grades above 69 and below 80
F	All grades below 70

Multi-Way If Statements Example

- Example:

```
number = int(input("Enter the numeric grade: "))
if number > 89:
        letter = 'A'
elif number > 79:
        letter = 'B'
elif number > 69:
        letter = 'C'
else:
        letter = 'F'
print("The letter grade is", letter)
```

Testing Selection Statements

- Tips:
  - Make sure that all of the possible branches or alternatives in a selection statement are exercised
  - After testing all of the actions, examine all of the conditions
  - Test conditions that contain compound Boolean expressions using data that produce all of the possible combinations of values of the operands

Random Numbers Example

- Programming languages include resources for generating **random numbers**
- **random** module supports several ways to do this
  - **randint** returns random number from among numbers between two arguments, included

```
import random
for roll in range(5):
	print(random.randint(1, 6))

2
4
6
4
3
```

Random Numbers Example

- Example: A simple guessing game

```
import random
computerNumber = random.randint(1,100)
count = 0
finished = False
while not finished:
    count += 1
    userNumber = int(input("Enter your guess: "))
    if userNumber < computerNumber:
        print("Too small!")
    elif userNumber > computerNumber:
        print("Too large!")
    else:
        print("Congratulations! You've got it in ", count, "tries!")
        finished = True
```

Random Numbers (3 of 3)

- Output from run:

Enter your guess: 50
Too small!
Enter your guess: 75
Too large!
Enter your guess: 63
Too small!
Enter your guess: 69
Too large!
Enter your guess: 66
Too large
Enter your guess: 65
You've got it in 6 tries!